

Base R: In-Class Exercise (Quiz)

Overview: We will modify the “big data” prison inmate count dataframe and then aggregate it to the state-level. Then merge it with census populations data to create an incarceration rate.

1. Load the “haven” package to import an SPSS file. If haven is not installed, please click on “packages” and “install” before loading the library.

```
library(haven)
```

#2. Create a dataframe named “prison_counts” by importing the SPSS (.sav) file “ncrp_3years_year_end.sav” from OneDrive (under the “Crime” and “Prison”) subfolders.

#3. Filter out unwanted years so that only data from the years 2000 and 2010 remains.

#4. Convert all variable names to lower case.

#5. Create a variable called “year” by assigning it the value of the rptyear variable.

#6. Create a dummy variable (inmate_dummy) that is equal to “1” on each row, to make inmate counting easier. Keep only the needed variables (year, dummy, state) using SUBSET function. **Also, create a counter for female inmates.

#7. Aggregate all needed variables using the AGGREGATE function. You may use CBIND to combine the female count with the overall inmate count (if you aggregated them separately).

#8. Rename or change the name of “inmate_dummy” to “inmate_count” using vector or bracket syntax [] and the NAMES function. Do the same for “female_count.”

#9. Import the census population data from OneDrive “census_panel.csv” (under the “Crime” subfolder).

#10. Merge the 2010 state-level inmate count data with census population data (using the Base R inner join function. Verify that the correct year has been merged.

```
#Example: database_inner <- merge(data_A, data_B, by = "ID")
```

#11. Create an incarceration rate (per 100,000 residents) for female inmates in each state.

#12. Remove “District of Columbia” (also known as Washington, DC) from the dataframe because this is a city, not a state.

#13. Sort the (from high to low incarceration_rate) using the order function, then print out the incarceration rate data with (including state name and year) so I can review all results.

```
Example: print(dataframe).
```

#14. Submit your code and final output via Blackboard.